Professional Issues in IT

SOFTWARE CONTRACTS

Software Contracts and Liability

What is a contract?

A contract is simply an agreement between two or more persons (the *parties* to the contract) that can be enforced in a court of law. The parties involved may be legal persons or natural persons.

Essentials for a contract

The most important and essential concerns of a contract are that:

- all the parties must intend to make a contract;
- all the parties must be competent to make a contract, that is, they must be old enough and of sufficiently sound mind to understand what they are doing;
- there must be a ‘consideration’, that is, each party must be receiving something and providing something.

- *Contract law governs the creation, execution, and enforcement of agreements between parties. It ensures that promises made in legally binding contracts are honored, and provides remedies if they are not. Contract law governs the creation, execution, and enforcement of agreements between parties. It ensures that promises made in legally binding contracts are honored, and provides remedies if they are not*

Licence Agreements

- When customers buy software, they are buying a copy of the software together with the right to use it in certain ways. The agreement that covers such purchases is a type of contract known as a **licence agreement.**
- ***For Example:***
  - *A licence may allow the licensees (i.e. the people to whom the software is licensed) to use one copy of the software on their computer. This is the type of licence commonly granted when software such as a computer game is purchased from a retail outlet.*
  - *A licence may allow licensees to run the software on a server on their local-area network and allow it to be used by any number of users simultaneously, up to some agreed maximum number. This type of licence is used, for example, for large multi-user database management systems and for applications such as accounting packages intended for the corporate market.*
  - *A licence may allow licensees to run as many copies of the software as they wish on computers at specified premises. This is often known as a* ***site licence.***

Licence Agreements

- There are several things that a software vendor may be concerned about when providing a customer with a licence to use one of its products:
  - making sure that it is not giving away any of its own rights in the software;
  - limiting the extent to which the customer can use the software, so that, if customers want to use it more extensively, they must pay an additional licence fee;
  - ensuring a regular income from support activities, possibly an annual maintenance charge, possibly fees for consultancy services;
  - ensuring that, as far as is possible, it will not be liable for any defects in the software.

Licence Agreements

- It is very common for the owner of the copyright in a piece of software to license other people or organisations to carry out some of the activities that are otherwise the exclusive right of the copyright owner. The copyright remains the property of the owner but the licensees acquire certain rights.

- Licences themselves may be granted in perpetuity (i.e. for ever) or for some fixed period. Licences for retail software are usually granted in perpetuity.
- Licences for corporate software, however, are typically granted on an annual basis; the initial fee may be larger than the fee for the annual renewal. Marketing licences are normally granted for a fixed period.

- The owner of the copyright can transfer ownership to someone else. This is known as assignment of the copyright and must be done in writing.

OUTSOURCING

- Outsourcing, sometimes known as facilities management, is the commercial arrangement under which a company or organisation (the customer) hands over the planning, management and operation of certain functions to another organisation (the supplier).
- The logic behind outsourcing is that a company that specialises in a particular area, be it IT or office cleaning, is likely to be able to make a better job of running services than an organisation whose main area of expertise is elsewhere.
- Forty years ago, companies would employ their own staff to develop software, run the staff canteen, clean the offices, carry out market research and so on. Nowadays, they are likely to outsource all these operations to specialist companies so that they can concentrate on their core business – that is, the business that is the main source of their **revenue.**

IT OUTSOURCING

- IT outsourcing contracts are inherently complex and depend very much on individual circumstances. The following is a list of just some of the points that need to be addressed:
  - how performance is to be monitored and managed;
  - what happens if performance is unsatisfactory;
  - which assets are being transferred;
  - staff transfers;
  - audit rights;
  - contingency planning and disaster recovery;
  - intellectual property rights in software developed during the contract;
  - duration of the agreement and termination provisions.

  The first two items above are key elements in IT outsourcing and are often treated as a separate annex to an agreement known as a service level annex.

FIXED PRICE CONTRACTS FOR BESPOKE SYSTEMS

- The first type of contract we shall consider is the type that is used when an organization is buying a system configured specifically to meet its needs. Such systems are known as ***tailor-made*** **or** ***bespoke*** **systems.**
- A bespoke system may consist of a single PC equipped with a word processor, a spreadsheet, and a set of macros adapted to the customer's needs or it may consist of several thousand PCs spread across 50 offices in different parts of the world, connected by a wide-area network, with large database servers and a million lines of specially written software.

Bespoke system

The contract for the supply of a bespoke system consists of three parts:

- A short *agreement*, which is signed by the parties to the contract: This states who the parties are and, very importantly, says that anything that may have been said or written before does not form part of the contract.
- The *standard terms and conditions*, which are normally those under which the supplier does business; and
- A set of *schedules* or *annexes*, which specify the particular requirements of this contract, including what is to be supplied, when it is to be supplied, what payments are to be made and when, and so on.

Issues in contract

- **What is to be produced**
- **What is to be delivered**
- **Ownership of rights**
- **Confidentiality**
- **Payment terms**
- **Penalty clauses**
- **Obligations of the client**
- **Standards and methods of working**

What is to be produced

- It is clearly necessary that the contract states what is to be produced.
- the standard terms and conditions refer to an annex and the annex then refers to a separate document that constitutes the **requirements specification.**
- It is important that the reference to the requirements specification identifies that document uniquely; normally this will mean quoting a date and issue number.

Problem: Any changes needed during the contract life

How are these changes to be accommodated by a contract that, in a sense, freezes the requirements of the parties to those at one particular time by incorporating the original specification into the contract?

What is to be delivered

The following is a non-exhaustive list of possibilities:

- source code;
- command files for building the executable code from the source and for installing it;
- documentation of the design and of the code;
- reference manuals, training manuals and operations manuals;
- software tools to help maintain the code;
- user training;
- training for the client's maintenance staff;
- test data and test results.

Ownership of rights

- software is potentially protectable by a number of intellectual property rights, such as copyright, design rights and trademarks, and rights of confidentiality may also apply to it.

- It is important that the contract should also state just what legal rights are being passed by the software house to the client under the contract.
- Ownership in physical items such as books, documents or disks will usually pass from the software house to the client, but other intangible rights, known as intellectual property rights, present more problems.

Confidentiality

- when a major bespoke software system is being developed, the two parties will acquire confidential information about each other.
- None of the parties would like the other to disclose its secrets.
- It is usual in these circumstances for each party to promise to maintain the confidentiality of the other's secrets, and for express terms to that effect to be included in the contract.

Payment terms

- The standard terms and conditions will specify the payment conditions –that is, something along the lines that:
- ***Payment shall become due within thirty days of the date of issue of an invoice. If payment is delayed by more than thirty days from the due date, the Company shall have the right, at its discretion, to terminate the contract, or to apply a surcharge at an interest rate of 2% above the bank base lending rate.***

Payment terms

A pattern of payments such as the following must be included:

- an initial payment of, say, 15 per cent of the contract value becomes due
- on signature of the contract;
- further stage payments become due at various points during the development,
- bringing the total up to, say, 65 per cent;
- a further 25 per cent becomes due on acceptance of the software;
- the final 10 per cent becomes due at the end of the warranty period.

Payment terms

- Such a pattern has advantages for the supplier in that it reduces the financial risk arising from possible insolvency of the client or from default for other reasons and it reduces possible cash flow difficulties
- In negotiating the payment pattern, the supplier will usually seek to have the stage payments becoming due on fixed calendar dates, while the client will try to have them tied to the achievement of specific project milestones, for example, approval of the design specification

Calculating payments for delays and changes

- Both sides of the parties may create little or high loss regarding unable of meeting the timeline. Thus extra work has to be indulged producing extra payments
- The contract must specify the process by which these extra payments are to be calculated.
- Delay payments and payments for variations to the original requirements are, perhaps, the commonest cause of contractual disputes, not only in software engineering but in most other contracting industries – the construction industry is a notorious example.

Penalty clauses

The previous subsection dealt with compensation for delays caused by the client; delays caused by the supplier are handled differently.

Delays in delivering working software are notoriously common; it might therefore be expected that contracts for the supply of software would normally include such a penalty clause. There are three reasons for this:

- Suppliers are very reluctant to accept penalty clauses and anything stronger than the example quoted above is likely to lead to reputable suppliers refusing to bid.
- If the contract is to include penalty clauses, the bid price is likely to be increased by at least half the maximum value of the penalty.
- If the software is seriously late and penalties approach their maximum, there is little incentive for the supplier to complete the work since they will already have received in stage payments as much as they are going to get.

Obligations of the client

- provide documentation on aspects of the client's activities or the environment in which the system will run;
- provide access to appropriate members of staff;
- provide machine facilities for development and testing;
- provide office accommodation and secretarial facilities for the
- company's staff when they are working on the client's premises;
- provide data communications facilities to the site.

The general terms and conditions will normally state that a list of specific obligations and the dates at which they will be required is given in an annex.

Standards and methods of working

- The supplier is likely to have company standards, methods of working, quality assurance procedures, and so on, and will normally prefer to use these. More sophisticated clients will have their own procedures and may require that these be adhered to.
- In some cases, the supplier may be required to allow the client to apply quality control procedures to the project.
- The contract must specify which is to apply.

Project meeting

- Regular progress meetings are essential to the successful completion of a fixed price contract and it is advisable that standard terms and conditions require them to be held. The minutes of progress meetings, duly approved and signed, should have contractual significance in that they constitute evidence that milestones have been reached (so that stage payments become due) and that delay payments have been agreed.

Acceptance procedures

- Acceptance procedures are a critical part of any fixed price contract for they provide the criteria by which successful completion of the contract is judged.
- The essence of the acceptance procedure is that the client should provide a fixed set of acceptance tests and expected results and that successful performance of these tests shall constitute acceptance of the system.
- The tests must be provided at or before the start of the acceptance procedure; within
- reason, there may be as many tests as the client wishes, but extra tests cannot be added once the test set has been handed over.
- The purpose of this restriction is to ensure that the acceptance procedure can be completed in reasonable time. Other points to be addressed under this heading include who shall be present when the tests are carried out and what happens if the tests are not completed successfully.

Warranty and maintenance

- Once the product has been accepted, it is common practice to offer a warranty period of, typically, 90 days. Any errors found in the software and reported within this period will be corrected free of charge. This clause is, of course, subject to negotiation; reducing or eliminating the warranty period will reduce the overall cost of the contract and prolonging the period will increase it.
- Once the warranty period is over, the supplier may offer, or the client
- demand, that maintenance will continue to be available on request.
- the resources required are unpredictable – the client almost certainly does not know what enhancements will be equired in two years' time.
- For this reason, a fixed price for the maintenance will not be appropriate. Maintenance will therefore usually be charged on a time and materials basis; the client may possibly be required to commit to taking a fixed number of days of effort each year in order to compensate the supplier for the need to retain knowledge of the system.

Inflation

- In lengthy projects or projects where there is a commitment to long-term maintenance, the supplier will wish to ensure protection against the effects of unpredictable inflation. To handle this problem, it is customary to include a clause which allows charges to be increased in accordance with the rise in costs.
- The clause should state how often (once a year, twice a year) charges can be increased and how the effect on the overall price is to be calculated.

Indemnity

- It could happen that, as a result of the client's instructions, the supplier is led unwittingly to infringe the intellectual property rights of a third party or that, through carelessness or dishonesty, the supplier provides a system which infringes such rights – perhaps through using proprietary software as a component of the system delivered.
- For this reason, it is advisable to include a clause under which each party indemnifies the other (that is, guarantees to cover any costs the other party becomes subject to) for liability arising from its own faults in this respect.

Termination of contract

- There are many reasons why it may become necessary to terminate a contract before it has been completed. It is not uncommon, for example, for the client to be taken over by another company that already has a system of the type being developed, or for a change in policy on the part of the client to mean that the system is no longer relevant to its needs.
- It is essential, therefore, that the contract make provision for terminating the work in an amicable manner. This usually means that the supplier is to be paid for all the work carried out up to the point where the contract is terminated, together with some compensation for the time needed to redeploy staff on other revenue earning work. The question of ownership of the work so far carried out must also be addressed.

Arbitration

- Contracts often therefore contain a clause saying that, in the event of a dispute that they cannot solve themselves, the parties agree to take the dispute first to **mediation** through a body such as the Centre for Effective Dispute Resolution. If that fails then the contract will specify whether disputes go to court or to **arbitration.**
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Arbitration

- An arbitration clause will usually state that, if arbitration is required, it will take place in accordance with the Arbitration Act 1996. This Act of Parliament lays down a set of rules for arbitration that cover many eventualities, and reference to it avoids the need to spell these out in detail; most of the provisions of the Act are optional, in the sense that they come into effect only if the contract contains no alternative provision.

CONSULTANCY AND CONTRACT HIRE

Consultancy

- Consultants are experts who are called in by an organisation to assess some aspect of its operations or its strategy and to make proposals for improvements. This means that the end product of a consultancy project is usually a report or other document.
- Under normal circumstances a fee for IT consulting is measured on a per day, per consultant basis.
- Fixed fee IT consulting contract applies to projects which are well defined.
- Open ended consultancy models generally favour the consulting firm, as the consultancy firm is rewarded on a per day basis, there is no incentive to complete assignments within a fixed time. The result often being risk of project and cost overrun.
- Contract is very simple.

There are four important aspects of a consultancy contract:

- Confidentiality: Consultants are often in a position to learn a lot about the companies for which they carry out assignments and may well be in a position to misuse this information for their own profit.
- Terms of reference: It is important that the contract refers explicitly to the terms of reference of the consultancy team and, in practice, these are perhaps the commonest source of disagreements in consultancy projects.
- Liability : Most consultants will wish to limit their liability for any loss that the customer suffers as a result of following their advice. Customers may not be happy to accept this and, in some cases, may insist on verifying that the consultant has adequate professional liability insurance.
- Control: It is common practice for the contract to require that a draft version of the final report be presented to the client. The client is given a fixed period to review the report and, possibly, ask for changes. The revised version that is then submitted by the consultant should be the final version.

Contract hire

- Supplier's responsibility is limited to providing suitably competent people and replacing them if they become unavailable.
- The staff work under the direction of the client.
- Payment is on the basis of a fixed rate for each day worked.
- Ownership of intellectual property rights generated in the course of the work may be needed to be addressed.

Time and Materials

- A time and materials contract (often referred to as a ‘cost plus’ contract) is somewhere between a contract hire agreement and a fixed price contract.
- The supplier agrees to undertake the development of the software in much the same way as in a fixed price contract, but payment is made on the basis of the costs incurred, with labour charged in the same way as for contract hire.
- The supplier is not committed to completing the work for a fixed price, although a maximum payment may be fixed beyond which the project may be reviewed.

HEALTH AND SAFETY

The ones that are of particular concern to software engineers are:

- provision and maintenance of safe plant;
- provision and maintenance of safe systems of work;
- provision of such information, instruction, training, and supervision as necessary;
- ensuring the workplace is maintained in a safe condition;
- provision and maintenance of a safe working environment and adequate welfare arrangements.

The Act also requires employers to ensure that their activities do not expose the general public to risks to their health and safety. Manufacturers of equipment to be used at work also have a responsibility to ensure that it is safe.

Failure to comply with the Health and Safety at Work Act is a criminal offence and, in serious cases, can lead to criminal proceedings being taken against individuals.